ESG Scores and Energy-Sector Relative Returns

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Research question

Do lagged ESG scores help predict one-year benchmark-relative returns, and how do energy company groups compare on risk-adjusted returns?

Data and methods

Twelve listed US and European energy companies; monthly observations spanning 2013–2024 across three political periods. Industry groups are utilities/transition and oil and gas.

Random Forest regressions with temporal evaluation, sector-relative returns, equal-weight portfolios and paired block-bootstrap confidence intervals.

Selected results

- In the updated evaluation, out-of-sample R^2 is negative in five of six region-period models.
- The single positive test is US 2017–2020 ($R^2 = 0.396$), based on only eight test months.
- EU 2017–2020 information-ratio difference: 1.676 (95% bootstrap interval 0.196–3.247).

Interpretation and limitations

Utility versus oil-and-gas membership confounds ESG with industry. Return construction, short test windows and multiple comparisons limit inference. The separate original 2025 report retains its historical figures.

Reproduce

```
python scripts/fetch_originals.py; python src/esg_analysis.py
```

Repository and full documentation